A Recall AUC, all mutations

B Recall AUC, de novo

C Recall AUC, gap/stop mask

D Recall AUC, de novo + gap/stop mask

E Max precision, all mutations, 1+ cases

F Max precision, de novo, 1+ cases

G Max precision, all mutations, 100+ cases

H Max precision, de novo, 100+ cases

antiGen
Where x which + PSSM, val−selected (10x)
antiGen, val−selected (10x)
Mean of z−scores
Mean of log−probs
Mean of ranks
antiGen as Tranception full FT
antiGen as EVE full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) LM head FT
BA.2 DMS (LY−CoV1404 escape)
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
Where × which
Substitution matrix
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
Codon table
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape

3rd
2nd
1st

0.5 0.6 0.7 0.8
AUC, recall by quantile

0.45 0.55 0.65 0.75 0.85
AUC, recall by quantile

0.00 0.25 0.50 0.75
Max PSSM precision

0.0 0.2 0.4 0.6
Max PSSM precision

0.0 0.1 0.2 0.3
Max PSSM precision

0.000 0.025 0.050 0.075 0.100
Max PSSM precision

Model type
antiGen
antiGen Finetunes
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Ensembles
Val−selected, averaged over 10 holdouts